

Fushuai Jiang

CONTACT INFORMATION	Department of Mathematics City University of Hong Kong Tat Chee Ave, Kowloon Tong Kowloon, Hong Kong	fusjiang@cityu.edu.hk https://jfsmath.github.io/
RESEARCH	Whitney extension theory, real & functional analysis, analysis on metric spaces	
ACADEMIC EMPLOYMENT	2026-Current 2023-2025 2022 Fall	City University of Hong Kong, Assistant Professor University of Maryland, Novikov Postdoctoral Fellow ICERM, Postdoctoral Fellow
EDUCATION	University of California, Davis Ph.D. in Mathematics, June 2022 Dissertation Topic: Nonnegative Smooth Interpolation Advisor: Kevin Luli University of California, Los Angeles B.S. in Mathematics, June 2017 Departmental Highest Honors, Cum Laude Latin Honors	
JOURNAL PUBLICATIONS	17. J. Chhoa, M. Ivanitskiy, F. Jiang, S. Li, D. McBride, T. Needham, and K. O'Hare Metric properties of partial and robust Gromov-Wasserstein distances Analysis and Geometry in Metric Spaces, 14(1):20250041, 2026 16. F. Jiang Roots, trace, and extendability of flat nonnegative functions International Mathematics Research Notices, online, 2024 15. F. Jiang, C. Liang, Y. Liang, and G.K. Luli Univariate Range-Restricted C^2 Interpolation Algorithms, Journal of Computational and Applied Mathematics, 425:115040, 2023. 14. C. Fefferman, F. Jiang, G.K. Luli C² interpolation with range restriction, Revista Matemática Iberoamericana, 39(2):649–710, 2023 13. F. Jiang, G.K. Luli, K. O'Neill Smooth selection for infinite sets, Advances in Mathematics, 407:108566, 2022. 12. F. Jiang, G.K. Luli, K. O'Neill, On the shape fields finiteness principle International Mathematics Research Notices, 23:18895–18918, 2022. 11. F. Jiang, G.K. Luli, Algorithms for nonnegative $C^2(\mathbb{R}^2)$ interpolation. Advances in Mathematics, 385:107756, 2021. 10. F. Jiang, G.K. Luli, C²($\mathbb{R}^2$) nonnegative extension by bounded-depth operators. Advances in Mathematics, 375:107391, 2020.	

9. F. Jiang, G.K. Luli,
[Nonnegative \$C^2\(\mathbb{R}^2\)\$ interpolation.](#)
Advances in Mathematics, 375:107364, 2020.
8. K. Xu, F. Jiang, W. Zhang, and Y. Hao
[Micromachined integrated self-adaptive nonlinear stops for mechanical shock protection of MEMS.](#)
Journal of Micromechanics and Microengineering, 28:064006, 2018.

PREPRINTS
&
CONFERENCE PROC.

7. F. Jiang
On a family of one-dimensional oscillation inequalities
[arXiv:2608.04639](#)
6. F. Jiang and G.K. Luli
Smooth solutions to systems of linear inequalities on $\mathbb{R}$
[arXiv:2606.22805](#)
5. F. Jiang and G.K. Luli
Optimal $C^{1,1}$ and Quasi-Optimal C^2 Monotone Interpolation with Curvature Control
[arXiv:2605.14302](#)
4. R. Balan, F. Jiang
Factorization of positive-semidefinite operators with absolutely summable entries
[arXiv:2409.20372](#).
3. F. Jiang
Nonnegative Whitney extension problem for $C^1(\mathbb{R}^n)$.
[arXiv:1912.06327](#).
2. K. Xu, N. Zhu, F. Jiang, W. Zhang, and Y. Hao
A Transfer Function Approach to Shock Duration Compensation for Laboratory Evaluation of Ultra-High-G Vacuum-Packaged MEMS Accelerometers.
IEEE 32nd International Conference on Micro Electro Mechanical Systems (MEMS), 676–679, 2019.
1. K. Xu, N. Zhu, F. Jiang, W. Zhang, and Y. Hao
Micromachined integrated shock protection via a self-adaptive nonlinear system.
19th International Conference on Solid-State Sensors, Actuators and Microsystems (TRANSDUCERS), 524–527, 2017.

HONORS AND
AWARDS

- | | |
|-----------|--|
| 2021-2022 | Yueh-Jing Lin Fund.
Awarded to high-achieving mathematics students. |
| 2021-2022 | Graduate Research Award.
Annual internal fellowship awarded to promising graduate researchers. |
| 2020 | Alice Leung Scholarship.
Awarded to graduate student with exceptional promise in mathematical research. |
| 2020-2021 | Summer Graduate Student Researcher (GSR) Award.
In recognition of exceptional achievements in our graduate education community. |

TALKS

01/09/2025	<i>Efficient algorithm for non-negative smooth interpolation</i> AMS Special Session on Adversarial, Interpretable, and Explainable AI
11/21/2024	<i>Finding a smooth solution to an underdetermined system</i> One World Mathematics of Information, Data, and Signals (1W-MINDS) Seminar
10/19/2024	<i>ℓ_1-squared summable series and a problem by Feichtinger</i> Frame Theory Days 2024
10/11/2024	<i>How to find a smooth solution to an underdetermined system?</i> Johns Hopkins postdoc seminar
07/06/2024	<i>Smooth selection of convex sets</i> Loo-Keng Hua Lecture , Chinese Academy of Sciences
06/24/2024	<i>Partial optimal transport</i> MRC 2024: Mathematics of Adversarial, Interpretable, and Explainable AI
03/24/2024	<i>Primal and dual optimization problems related to matrix factorizations</i> AMS Special Session on Bases and Frames in Hilbert Spaces, III
08/14/2023	<i>Positivity and the Whitney Extension Problem</i> The 15th Whitney Problems Workshop
06/18/2023	<i>Recent development in Whitney extension problems with constraint</i> Shenzhen University
03/10/2023	<i>Quasi-optimal C^2 interpolation with range restriction</i> CUNY GC Harmonic Analysis and PDE Seminar
10/14/2022	<i>Smooth Selection of Convex Sets</i> Harmonic Analysis and Convexity Program , ICERM
10/06/2022	<i>Quasi-optimal C^2 interpolation with range restriction</i> Fall Fourier Talks, University of Maryland at College Park
07/08/2021	<i>Recent Results on Whitney extensions with constraints</i> The 14th Whitney Problems Workshop
05/15/2020	<i>Efficient algorithm for nonnegative C^2 interpolation</i> UC Davis PDE and Applied Math Seminar
02/23/2020	<i>A gentle introduction to pseudodifferential calculus</i> UC Davis Student-run Analysis & PDE Seminar
10/15/2019	<i>Fitting smooth functions to data - Whitney's problems and beyond</i> UC Davis Student-run Research Seminar
05/11/2019	<i>Whitney extension problem and interpolation of data</i> Bay Graduate Math Conference
02/08/2019	<i>Interpolation of data by nonnegative C^2 functions</i> UC Davis PDE and Applied Math Seminar

TEACHING	Fall	2025	Data Visualization and Presentation (R)
	Spring	2025	Applied Probability and Statistics (senior level, R)
	Fall	2024	Data Visualization and Presentation (R)
	Spring	2024	Data Science and Machine Learning (Python)
	Spring	2024	Applied Probability and Statistics (senior-level, R)
	Fall	2023	Calculus III
	Spring	2023	Linear Algebra and Applications (MATLAB)
	Summer	2022	Real Analysis
	Fall	2020	Teaching Assistant, Real Analysis
	Spring	2020	Teaching Assistant, Graduate Analysis
	Summer	2019	Instructor, Calculus II
	Spring	2019	Teaching Assistant, Graduate Analysis
	Various	2017-2021	Teaching Assistant, Calculus I-III
SERVICE	2026		<i>Faculty supervisor</i> , CityUHK IAS student internship
	2024		<i>Assistant organizer</i> , Mathematics Research Communities, AMS
	2021-now		<i>Mentor</i> , Undergraduate research
	2023		<i>Organizer</i> , The 15th Whitney Problems workshop
	2021		<i>Organizer</i> , The 14th Whitney Problems workshop
	2021		<i>Organizer</i> , 2021 Davis Math Conference
	2021		<i>Organizer</i> , Student Run Analysis&PDE Seminar
	2021-2022		<i>Mentor</i> , Direct Reading Project
	2018-2020		<i>Tutor</i> , STEM Café at Women's Resources and Research Center
	2019-2020		<i>Vice President</i> , Galois group (graduate student group)
	2018-2020		<i>Representative</i> , Graduate Student Association
PROGRAMMING SKILLS	R		(dplyr, ggplot, Shiny)
	Python		(NumPy, SciPy, scikit-learn, pandas)
	MATLAB		